Status	Finished
Started	Sunday, 25 October 2026, 4:01 PM
Completed	Sunday, 25 October 2026, 4:07 PM
Duration	6 mins 2 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

a. It divides the list into sublists
b. It uses the heap data structure
c. It bubbles up the largest element in each pass
d. It recursively sorts subproblems

The correct answer is: It bubbles up the largest element in each pass

Question **2**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Removing duplicates from a list
b. Finding a specific element in a list
c. Arranging data in a specific order
d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **3**

Complete

Mark 1.00 out of 1.00

Which of the following is a key reason for the importance of sorting algorithms?

a. Sorting is rarely used in programming
b. Sorting makes it harder to search for items
c. Sorting decreases the efficiency of selection operations
d. Sorting helps in finding duplicates quickly

The correct answer is: Sorting helps in finding duplicates quickly

Question **4**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Quick Sort
b. Bubble Sort
c. Merge Sort
d. Insertion Sort

The correct answer is: Bubble Sort

Question **5**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Sorting the entire list sequentially
b. Comparing each element with the others
c. Combining sorted sublists
d. Dividing the input into smaller subproblems

The correct answer is: Dividing the input into smaller subproblems

Question **6**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

a. It has no effect on the search operation
b. It makes the search operation slower
c. It increases the number of comparisons needed
d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **7**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

a. They are compared element by element
b. They are split again into smaller sublists
c. They are discarded
d. They are combined to form a single sorted list

The correct answer is: They are combined to form a single sorted list

Question **8**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

a. It makes the analysis slower
b. It allows for quicker identification of duplicates
c. It has no effect on the analysis process
d. It complicates the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **9**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

a. It repeatedly finds the minimum element and moves it to the sorted part of the list
b. It compares adjacent elements and swaps them if necessary
c. It divides the list into two halves, sorts each half, and then merges them
d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **10**

Complete

Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

a. It does not require recursion
b. It sorts data using a single pass
c. It is based on the divide-and-conquer approach
d. It is less efficient than bubble sort

The correct answer is: It is based on the divide-and-conquer approach

Question **11**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

a. Complexity
b. Sorting
c. Searching
d. Merging

The correct answer is:
Complexity

Question **12**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Selection
- b. Duplicates
- c. All of the above
- d. Searching

The correct answer is: All of the above

Question **13**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Bubbling up the largest element to its correct position with each pass
- b. Being the most efficient sorting algorithm
- c. Using the divide-and-conquer approach
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **14**

Complete

Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Bubbling up the smallest element
- b. Dividing the list into halves
- c. Bubbling up the largest element to its correct position
- d. Sorting the entire list in one pass

The correct answer is: Bubbling up the largest element to its correct position

Question **15**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

a. Inserting data into a list
b. Deleting data from a list
c. Searching for data in a list
d. Arranging data in a particular format

The correct answer is: Arranging data in a particular format